

ATAC-seq Analysis Pipeline

Bash implementation -> Nextflow workflow

Paired-end ATAC-seq | hg38 | MACS2 peak calling

ERR154875

29

Example paired-end
dataset

hg38

Human reference genome

7 stages

QC to final reporting

32,738

MACS2 peaks identified

Project documentation derived from the implemented Bash / Nextflow workflows and validation outputs.

Pipeline Workflow

Seven major stages from paired-end FASTQ reads to quality reporting.



Core analysis tools

FastQC | Trim Galore | Bowtie2 | SAMtools | MACS2 | DeepTools | MultiQC

Environment & Input Data

Execution context and files used for the example analysis.

Computational environment

- WSL Ubuntu
- Conda environment: atacseq_final
- Four threads configured for processing
- Human hg38 reference genome
- Bowtie2 alignment index configured for hg38

Input data

- Sample accession: ERR15487529
- Paired-end FASTQ: ERR15487529_1.fastq.gz
- Paired-end FASTQ: ERR15487529_2.fastq.gz
- Reference: hg38
- Source accession documented in the project README

Implementation & Reproducibility

The repository preserves the original Bash workflow and its Nextflow conversion.

Bash implementation

- End-to-end command-line workflow
- FastQC and Trim Galore
- Bowtie2 alignment to hg38
- SAMtools filtering and indexing
- MACS2 peak calling
- DeepTools coverage generation
- MultiQC quality reporting

Nextflow implementation

- Converted from the Bash workflow
- Seven modular processes
- Paired-end file handling with channels
- Process-level output publishing
- Parameter-based configuration
- Improved workflow organization and reproducibility

```
Run: nextflow run nextflow/atac_seq_pipeline.nf
```

Outputs & Validation

Key results from the processed sample, with genome-browser visualization.

89.8%

Overall Bowtie2 alignment rate

32,738

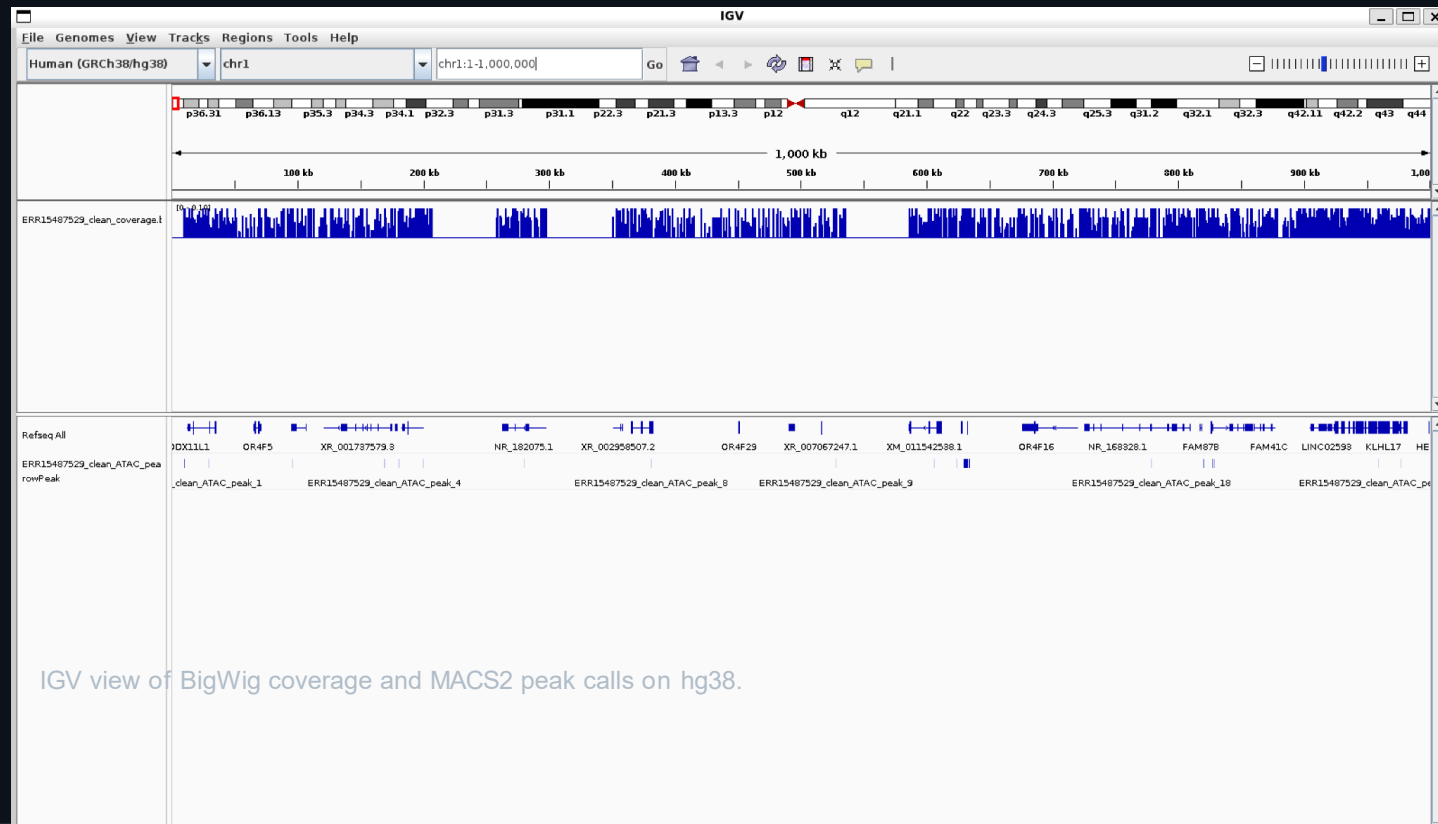
MACS2 peaks

124 bp

MACS2 fragment length

~8.9 M

Reads per mate



Generated outputs

- Filtered BAM + index
- MACS2 peak files
- BigWig coverage track
- MultiQC HTML report

IGV view of BigWig coverage and MACS2 peak calls on hg38.

ATAC-seq Applications

The workflow supports downstream chromatin-accessibility and regulatory analyses.

Chromatin accessibility

Identify open chromatin regions

Regulatory elements

Explore promoters and enhancers

Comparative analysis

Compare accessibility across conditions

Cell profiling

Extend to single-cell ATAC-seq

Multi-omics integration

Integrate with RNA-seq / ChIP-seq

Disease research

Support biomarker-oriented studies

Documentation prepared for the GitHub project repository; raw sequencing data and large intermediate files are not included.